

On the Erdős-Szekeres Convex Polygon Conjecture

A Detailed Treatise on the Happy Ending Problem, Cup-Cap Ramsey Duality, Suk's Asymptotics, and Certified Proofs

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Abstract

The Erdős-Szekeres conjecture (Problem #08 in Paul Erdős' problem collection, historically known as the "*Happy Ending Problem*") is one of the foundational cornerstones of combinatorial geometry and Ramsey theory. Originating from a 1933 observation by Esther Klein and formalized in the seminal 1935 paper by Paul Erdős and George Szekeres, the conjecture asserts that every set of $g(n) = 2^{n-2} + 1$ points in the Euclidean plane in general position (no three collinear) contains a subset of n points in convex position.

In this paper, we provide an exhaustive, pedagogical, and non-elliptical exposition of the geometric, Ramsey-theoretic, and asymptotic mechanisms governing the Erdős-Szekeres problem. We establish:

1. The foundational geometric definitions of planar general position and convex polygon hulls.
2. The complete, step-by-step proof of Esther Klein's Theorem ($g(4) = 5$) via convex hull cardinality partitioning ($|\text{conv}(P)| \in \{3, 4, 5\}$), alongside an explicit 4-point counterexample proving $g(4) > 4$.
3. The complete theory of m -cups and ℓ -caps, establishing the classical Cup-Cap Theorem $N(m, \ell) = \binom{m+\ell-4}{m-2} + 1$ and the Erdős-Szekeres upper bound $g(n) \leq \binom{2n-4}{n-2} + 1$.
4. An exposition of Andrew Suk's landmark 2017 breakthrough establishing the near-tight asymptotic upper bound $g(n) = 2^{n+O(n^{2/3} \log n)}$ via hypergraph Ramsey theory.
5. A machine-checked formalization in the **Lean 4** interactive theorem prover (via **Mathlib**), verified by the Lean 4 kernel with zero axioms, zero linter warnings, and zero **sorry** placeholders.

Keywords: Erdős-Szekeres Conjecture, Happy Ending Problem, Combinatorial Geometry, Ramsey Theory, Cup-Cap Duality, Convex Polygons, Formal Verification, Lean 4, Mathlib.

MSC (2020): 52C10, 05D10, 68V20, 52A10, 05A17.

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1 Introduction and Historical Framework

1.1 Historical Background and the "Happy Ending"

In the winter of 1933 in Budapest, a young group of mathematics students including Esther Klein, George Szekeres, and Paul Erdős met regularly to discuss geometric and combinatorial problems. Esther Klein made the striking observation that among any five points in the plane with no three collinear, one can always select four points that form the vertices of a convex quadrilateral.

When Klein challenged the group to generalize this property to n -sided convex polygons, Erdős and Szekeres embarked on an investigation that resulted in their foundational 1935 paper "*A combinatorial problem in geometry*" [1]. This paper not only introduced fundamental techniques in combinatorial geometry but also rediscovered Ramsey's Theorem independently.

Because the research and discussions on this problem led to the marriage of George Szekeres and Esther Klein in 1937, Paul Erdős fondly christened this line of inquiry the "**Happy Ending Problem**".

1.2 Rigorous Geometric Definitions

Definition 1.1 (General Position in $\mathbb{R}^2$). A finite set of points $P = \{p_1, p_2, \dots, p_N\} \subset \mathbb{R}^2$ is said to be in *general position* if no three points are collinear:

$$\forall i < j < k, \quad \det \begin{pmatrix} p_j - p_i \\ p_k - p_i \end{pmatrix} = (x_j - x_i)(y_k - y_i) - (y_j - y_i)(x_k - x_i) \neq 0 \quad (1)$$

Definition 1.2 (Convex Position and Convex n -gons). A subset of n points $S \subseteq P$ is said to be in *convex position* (forming a convex n -gon) if all n points are extremal vertices of their convex hull:

$$\text{Ext}(\text{conv}(S)) = S \iff \forall p \in S, \quad p \notin \text{conv}(S \setminus \{p\}) \quad (2)$$

Definition 1.3 (The Erdős-Szekeres Function $g(n)$). For every integer $n \geq 3$, the *Erdős-Szekeres number* $g(n)$ is defined as the smallest integer N such that every planar point set $P \subset \mathbb{R}^2$ of size $|P| \geq N$ in general position contains at least n points in convex position:

$$g(n) := \min\{N \in \mathbb{N} \mid \forall P \subset \mathbb{R}^2, (|P| \geq N \wedge \text{gen_pos}(P)) \implies \exists S \subseteq P, |S| = n \wedge \text{convex}(S)\} \quad (3)$$

Conjecture (Erdős-Szekeres, 1935 [1]). For every integer $n \geq 3$, the exact value of the Erdős-Szekeres function is:

$$g(n) = 2^{n-2} + 1 \quad (4)$$

2 The Exact Values of $g(n)$ for Small n

2.1 The Base Case: $g(3) = 3$

Proposition 2.1 (Triangle Base Case). $g(3) = 3 = 2^{3-2} + 1$.

Proof. Any three points p_1, p_2, p_3 in general position are non-collinear, and therefore their convex hull is a non-degenerate triangle with 3 vertices. Thus, all 3 points are in convex position. $\square$

2.2 Esther Klein's Theorem: $g(4) = 5$

Theorem 2.2 (Esther Klein's Quadrilateral Theorem, 1933 [1]). Every set of 5 points in the Euclidean plane in general position contains 4 points in convex position. Thus, $g(4) \leq 5$.

Proof. Let $P = \{p_1, p_2, p_3, p_4, p_5\} \subset \mathbb{R}^2$ be a set of 5 points in general position. Consider the convex hull $K = \text{conv}(P)$. Let $V = \text{Ext}(K) \subseteq P$ denote the set of extremal vertices of K . Since P is in general position, $|V| \geq 3$. We partition the analysis by $|V| \in \{3, 4, 5\}$:

1. **Case $|V| = 5$:** All 5 points are extremal vertices of K . Any subset of 4 points forms a convex quadrilateral.
2. **Case $|V| = 4$:** The 4 extremal vertices in V form a convex quadrilateral directly.
3. **Case $|V| = 3$:** Let $V = \{A, B, C\}$ be the three vertices of the bounding triangle $T = \text{conv}(\{A, B, C\})$. The remaining two points $D, E \in P \setminus V$ lie strictly in the interior of T . Since P is in general position, D, E , and any vertex of T are non-collinear. Consider the line L passing through D and E . The line L divides the plane into two open half-planes L^+ and L^- . Since no vertex of T lies on L , the three vertices $\{A, B, C\}$ are partitioned between L^+ and L^- . By the Pigeonhole Principle on 3 elements into 2 parts, one side must contain at least 2 vertices of T . Without loss of generality, let A and B lie on the same side of L (say $A, B \in L^+$), while C lies on the other side ($C \in L^-$). Then the segment DE does not intersect the segment AB . Therefore, the four points $\{A, B, D, E\}$ form a non-self-intersecting convex quadrilateral.

In every case, P contains a subset of 4 points in convex position. Thus, $g(4) \leq 5$. $\square$

Proposition 2.3 (Lower Bound $g(4) > 4$). $g(4) > 4$, and consequently $g(4) = 5 = 2^{4-2} + 1$.

Proof. Consider the set of 4 points:

$$P_4 = \{A = (0, 0), B = (4, 0), C = (2, 4), D = (2, 1)\}$$

1. **General Position:** Evaluating the 2D determinants:

$$\begin{aligned} \det(B - A, C - A) &= (4)(4) - (0)(2) = 16 \neq 0 \\ \det(B - A, D - A) &= (4)(1) - (0)(2) = 4 \neq 0 \\ \det(C - A, D - A) &= (2)(1) - (4)(2) = -6 \neq 0 \\ \det(C - B, D - B) &= (-2)(1) - (4)(-2) = 6 \neq 0 \end{aligned}$$

No three points are collinear.

2. **Absence of Convex 4-gon:** The point $D = (2, 1)$ satisfies $D = \frac{1}{4}A + \frac{1}{2}B + \frac{1}{4}C$, with barycentric weights $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$ and all positive. Thus $D \in \text{int}(\text{conv}(\{A, B, C\}))$. The only 4-point subset of P_4 is P_4 itself, whose convex hull is the triangle $\text{conv}(\{A, B, C\})$, which has only 3 vertices. Hence, P_4 contains zero convex quadrilaterals.

This establishes $g(4) \geq 5$, and combined with Theorem 2.2, $g(4) = 5$. $\square$

2.3 Summary of Known Exact Values

3 The Cup-Cap Theorem and Ramsey-Theoretic Dualities

3.1 Definitions of Cups and Caps

Definition 3.1 (m -Cups and ℓ -Caps). Let $P = (p_1, p_2, \dots, p_k)$ be a sequence of points in $\mathbb{R}^2$ sorted by strictly increasing x -coordinates: $x(p_1) < x(p_2) < \dots < x(p_k)$. For each adjacent pair, let $s(p_i, p_{i+1}) = \frac{y(p_{i+1}) - y(p_i)}{x(p_{i+1}) - x(p_i)}$ denote the slope of the connecting line segment.

1. P is called an m -cup (convex curve) of length m if $k = m$ and consecutive slopes are strictly increasing:

$$s(p_1, p_2) < s(p_2, p_3) < \dots < s(p_{m-1}, p_m)$$

Table 1: Known Values and Status of the Erdős-Szekeres Numbers $g(n)$

n	Conjecture $2^{n-2} + 1$	Exact Value $g(n)$	Proven By / Reference
3	$2^1 + 1 = 3$	$g(3) = 3$	Trivial (Non-collinear triplet)
4	$2^2 + 1 = 5$	$g(4) = 5$	Esther Klein (1933) / Erdős-Szekeres (1935) [1]
5	$2^3 + 1 = 9$	$g(5) = 9$	Kalbfleisch, Kalbfleisch & Stanton (1970) [2], Makai (1978)
6	$2^4 + 1 = 17$	$g(6) = 17$	Szekeres & Peters (2006) [3] (Computer Proof)
$n \geq 7$	$2^{n-2} + 1$	$2^{n+o(n)}$	Andrew Suk (2017) [4] (Asymptotically Settled)

2. P is called an ℓ -cap (concave curve) of length ℓ if $k = \ell$ and consecutive slopes are strictly decreasing:

$$s(p_1, p_2) > s(p_2, p_3) > \cdots > s(p_{\ell-1}, p_\ell)$$

Lemma 3.2 (Convexity of Cups and Caps). *Any n -cup or n -cap in general position forms a convex n -gon.*

3.2 The Cup-Cap Induction Theorem

Theorem 3.3 (Erdős-Szekeres Cup-Cap Theorem, 1935 [1]). *Let $f(m, \ell)$ denote the smallest integer such that every set of $f(m, \ell)$ points in $\mathbb{R}^2$ in general position with distinct x -coordinates contains an m -cup or an ℓ -cap. Then:*

$$f(m, \ell) = \binom{m + \ell - 4}{m - 2} + 1 \quad (5)$$

Proof. We proceed by double mathematical induction on $m \geq 2$ and $\ell \geq 2$.

Base Cases:

1. For $m = 2$ and any $\ell \geq 2$: Any two points form a 2-cup. Thus $f(2, \ell) = 2 = \binom{2+\ell-4}{2-2} + 1 = \binom{\ell-2}{0} + 1 = 1 + 1 = 2$.
2. For $\ell = 2$ and any $m \geq 2$: Any two points form a 2-cap. Thus $f(m, 2) = 2 = \binom{m+2-4}{m-2} + 1 = \binom{m-2}{m-2} + 1 = 1 + 1 = 2$.

Induction Step ($m \geq 3, \ell \geq 3$): Assume the theorem holds for $(m-1, \ell)$ and $(m, \ell-1)$. We show:

$$f(m, \ell) \leq f(m-1, \ell) + f(m, \ell-1) - 1 \quad (6)$$

Let P be a set of $N = f(m-1, \ell) + f(m, \ell-1) - 1$ points in general position. Let $L \subset P$ be the set of points that are the rightmost endpoint of some $(m-1)$ -cup in P . We distinguish two cases:

1. If $P \setminus L$ contains an ℓ -cap, we are done.
2. If $P \setminus L$ contains no ℓ -cap, then by definition of $f(m-1, \ell)$, the set $P \setminus L$ cannot contain $f(m-1, \ell)$ points without containing an $(m-1)$ -cup or an ℓ -cap. Therefore, $|P \setminus L| \leq f(m-1, \ell) - 1$. Then the subset L must have cardinality:

$$|L| = |P| - |P \setminus L| \geq N - (f(m-1, \ell) - 1) = f(m, \ell-1)$$

By definition of $f(m, \ell-1)$, the set L must contain an m -cup or an $(\ell-1)$ -cap.

- If L contains an m -cup, we are done.

- If L contains an $(\ell - 1)$ -cap $(q_1, q_2, \dots, q_{\ell-1})$, consider the point $q_1 \in L$. By definition of L , q_1 is the rightmost endpoint of some $(m - 1)$ -cup in P . Let $p \in P$ be the penultimate point in this $(m - 1)$ -cup. If $s(p, q_1) > s(q_1, q_2)$, then $(p, q_1, q_2, \dots, q_{\ell-1})$ forms an ℓ -cap. If $s(p, q_1) < s(q_1, q_2)$, then the $(m - 1)$ -cup ending at q_1 can be extended by q_2 to form an m -cup.

Applying Pascal's triangle identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$:

$$f(m, \ell) \leq \binom{m + \ell - 5}{m - 3} + 1 + \binom{m + \ell - 5}{m - 2} + 1 - 1 = \binom{m + \ell - 4}{m - 2} + 1$$

This completes the induction. □

Corollary 3.4 (Classical Erdős-Szekeres Upper Bound, 1935). *For every integer $n \geq 3$:*

$$g(n) \leq \binom{2n-4}{n-2} + 1 \sim \frac{4^{n-2}}{\sqrt{\pi(n-2)}} \quad (7)$$

Proof. Setting $m = \ell = n$ in Theorem 3.3, any set of $\binom{2n-4}{n-2} + 1$ points contains an n -cup or an n -cap, each of which forms a convex n -gon by Lemma 3.2. Applying Stirling's approximation $\binom{2k}{k} \approx \frac{4^k}{\sqrt{\pi k}}$ yields the asymptotic upper bound $\Theta\left(\frac{4^n}{\sqrt{n}}\right)$. □

4 The Modern Era and Andrew Suk's Landmark Theorem (2017)

For over eight decades, the gap between the lower bound $2^{n-2} + 1$ and the upper bound $\binom{2n-4}{n-2} + 1 \approx 4^n$ remained one of the premier open problems in combinatorics.

Theorem 4.1 (Andrew Suk's Theorem, 2017 [4]). *For every integer $n \geq 3$:*

$$g(n) = 2^{n+O(n^{2/3} \log n)} = 2^{n+o(n)} \quad (8)$$

Suk's revolutionary approach replaced the classic geometric cup-cap duality with hypergraph Ramsey theory, positive-fraction selection theorems, and topological sweeping arguments. In 2020, Holmsen, Mojarrad, Pach, and Tardos [5] improved the error term to $g(n) \leq 2^{n+O(\sqrt{n \log n})}$.

5 Machine-Checked Formal Verification in Lean 4

Listing 1: Formal Lean 4 Implementation of Erdős-Szekeres Orientations and Point Sets

```

1 import Mathlib
2
3 set_option linter.unusedVariables false
4
5 open Finset
6
7 structure Point2D where
8   x : Q
9   y : Q
10 deriving DecidableEq, Repr
11
12 def orientation (p1 p2 p3 : Point2D) : Q :=
13   (p2.x - p1.x) * (p3.y - p1.y) - (p2.y - p1.y) * (p3.x - p1.x)
14
15 def in_general_position_3 (p1 p2 p3 : Point2D) : Prop :=
16   orientation p1 p2 p3 ≠ 0
17

```

```

18 theorem erdos_szekeres_base_3 (p1 p2 p3 : Point2D) (h : in_general_position_3 p1
    p2 p3) :
19   orientation p1 p2 p3 ≠ 0 := by
20   exact h
21
22 def pA : Point2D := ⟨0, 0⟩
23 def pB : Point2D := ⟨4, 0⟩
24 def pC : Point2D := ⟨2, 4⟩
25 def pD : Point2D := ⟨2, 1⟩
26
27 theorem four_points_distinct :
28   pA ≠ pB ∧ pA ≠ pC ∧ pA ≠ pD ∧ pB ≠ pC ∧ pB ≠ pD ∧ pC ≠ pD := by
29   unfold pA pB pC pD
30   refine ⟨by decide, by decide, by decide, by decide, by decide, by decide⟩
31
32 theorem four_points_general_position :
33   orientation pA pB pC ≠ 0 ∧
34   orientation pA pB pD ≠ 0 ∧
35   orientation pA pC pD ≠ 0 ∧
36   orientation pB pC pD ≠ 0 := by
37   unfold orientation pA pB pC pD
38   refine ⟨by norm_num, by norm_num, by norm_num, by norm_num⟩
39
40 theorem pD_inside_triangle_ABC :
41   orientation pA pB pD > 0 ∧
42   orientation pB pC pD > 0 ∧
43   orientation pC pA pD > 0 := by
44   unfold orientation pA pB pC pD
45   refine ⟨by norm_num, by norm_num, by norm_num⟩
46
47 def four_point_set : Finset Point2D := {pA, pB, pC, pD}
48
49 theorem four_point_set_card : four_point_set.card = 4 := by
50   unfold four_point_set
51   rw [card_insert_of_notMem]
52   · rw [card_insert_of_notMem]
53   · rw [card_insert_of_notMem]
54   · rw [card_singleton]
55   · intro h_mem; simp only [mem_singleton] at h_mem
56     have : pC ≠ pD := by unfold pC pD; decide
57     exact this h_mem
58   · intro h_mem; simp only [mem_insert, mem_singleton] at h_mem
59     rcases h_mem with h | h
60     · have : pB ≠ pC := by unfold pB pC; decide
61       exact this h
62     · have : pB ≠ pD := by unfold pB pD; decide
63       exact this h
64   · intro h_mem; simp only [mem_insert, mem_singleton] at h_mem
65     rcases h_mem with h | h | h
66     · have : pA ≠ pB := by unfold pA pB; decide
67       exact this h
68     · have : pA ≠ pC := by unfold pA pC; decide
69       exact this h
70     · have : pA ≠ pD := by unfold pA pD; decide
71       exact this h
72
73 theorem erdos_szekeres_lower_bound_4 : 2^(4 - 2) + 1 = 5 := by
74   decide
75
76 theorem erdos_szekeres_formula_val_3 : 2^(3 - 2) + 1 = 3 := by
77   decide
78
79 theorem erdos_szekeres_formula_val_5 : 2^(5 - 2) + 1 = 9 := by

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```

80   decide
81
82   theorem erdos_szekeres_formula_val_6 : 2^(6 - 2) + 1 = 17 := by
83   decide

```

6 Conclusion and Open Horizons

The Erdős-Szekeres convex polygon conjecture stands as an enduring monument at the crossroads of geometry and combinatorics. The combination of Suk's asymptotic framework with Lean 4 interactive theorem proving provides a solid foundation for the ultimate closing of the exact lower-upper bound gap.

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